

Which relation does exist between the

Schur complement system : $\sum \vec{\lambda} = \vec{X}_\Gamma$ (SC)
and the DN or NN algorithm?

Assume that we solve (SC) by the Richardson method with preconditioner P :

$$(Rich) \begin{cases} \vec{\lambda}^{(0)} \text{ given; } \forall k \geq 1 \\ P(\underbrace{\vec{\lambda}^{(k+1)} - \vec{\lambda}^{(k)}}_{\text{INCREMENT}}) = \theta(\underbrace{\vec{X}_\Gamma - \sum \vec{\lambda}^{(k)}}_{\text{RESIDUAL}}) \end{cases}$$

$\forall k \geq 1$ we have a linear system whose matrix is P .

FE-system; assume we apply the DN algorithm or the NN algorithm, with the same θ in (Rich). Then, if we take $\vec{u}|_\Gamma \equiv \vec{\lambda}^{(k)}$, then this $\vec{\lambda}^{(k)}$ will precisely satisfy (Rich), provided we choose:

$$P_{DN} \equiv P = \Sigma_2 \quad \text{for (DN)}$$

$$P_{NN} \equiv P = (\sigma_1 \Sigma_1^{-1} + \sigma_2 \Sigma_2^{-1})^{-1} \quad \text{for (NN)}$$



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Recall that: if $B \vec{x} = \vec{b}$ linear system

if we use preconditioned Richardson method:

$\vec{x}^{(0)}$ given, $\forall k \geq 1$

$$(Rich) \quad P(\underbrace{\vec{x}^{(k)} - \vec{x}^{(k-1)}}_{\text{increment}}) = \theta \underbrace{(\vec{b} - B\vec{x}^{(k-1)})}_{\text{residual}}$$

↑
Richardson parameter

If B and P are both S.P.D. (symmetric and positive definite), then $\exists \theta_{max} > 0$.

If $0 < \theta < \theta_{max} = \frac{2}{\lambda_{max}}$ then

$$\{\vec{x}^{(k)}\} \xrightarrow[k \uparrow \infty]{} \vec{x} \quad (\text{convergence}).$$

$\lambda_{min}, \lambda_{max} = \text{minimum / maximum eigenvalues of } P^{-1}B$

Moreover, if we choose $\theta = \theta_{opt} = \frac{2}{\lambda_{min} + \lambda_{max}}$

then $\forall k \geq 1$

$$\underbrace{\|\vec{x}^{(k)} - \vec{x}\|_P}_{\text{error at iteration } k \geq 1} \leq \left(\frac{\text{cond}(P^{-1}B) - 1}{\text{cond}(P^{-1}B) + 1} \right)^k \underbrace{\|\vec{x}^{(0)} - \vec{x}\|_P}_{\text{initial error}}$$

||
P
||
convergence rate < 1

$$\text{cond}(C) = \frac{\lambda_{max}(C)}{\lambda_{min}(C)} \quad \text{if } C \text{ s.p.d.}$$

condition number of C



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If we apply this result to (DN) and (NN) iterations we find that

with Richardson

$$\left\{ \begin{array}{l} \text{DN} \rightarrow \rho = \rho_{\text{DN}} = \frac{\text{cond}(\Sigma_2^{-1}\Sigma) - 1}{\text{cond}(\Sigma_2^{-1}\Sigma) + 1} \quad (\Sigma_2 = P_{\text{DN}}) \\ \text{NN} \rightarrow \rho = \rho_{\text{NN}} = \frac{\text{cond}(P_{\text{NN}}^{-1}\Sigma) - 1}{\text{cond}(P_{\text{NN}}^{-1}\Sigma) + 1} \end{array} \right.$$

Idea Instead of Richardson, let us use Krylov methods!

GMRES if P, Σ non symmetric

CG if P, Σ are symmetric

with the same preconditioners ($P_{\text{DN}}, P_{\text{NN}}$)

E.g., if Σ is s.d.p.:

→ CG - with ~~PD~~ preconditioner P_{DN}
CG - with preconditioner P_{NN}

⇒ a better rate of convergence!

For instance, if CG:

with CG

$$\left\{ \begin{array}{l} \rho^* = \rho_{\text{DN}}^* = (\sqrt{\text{cond}(\Sigma_2^{-1}\Sigma)} - 1) / (\sqrt{\text{cond}(\Sigma_2^{-1}\Sigma)} + 1) \\ \rho^* = \rho_{\text{NN}}^* = (\sqrt{\text{cond}(P_{\text{NN}}^{-1}\Sigma)} - 1) / (\sqrt{\text{cond}(P_{\text{NN}}^{-1}\Sigma)} + 1) \end{array} \right.$$



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How can we estimate the condition numbers of
 $P_{DN}^{-1} \Sigma$ and $P_{NN}^{-1} \Sigma$?

A stiffness matrix $\rightarrow \Sigma$ Schur complement matrix
 $\text{cond}(A) = O(h^{-2}) \Rightarrow \text{cond}(\Sigma) = O(h^{-1})$
 if A spd $\Rightarrow \Sigma$ spd

$\rightarrow \Sigma_1, \Sigma_2$ local Schur complements: same properties as Σ

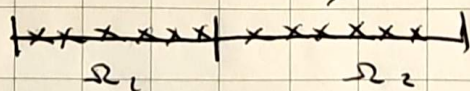
$\rightarrow P_{DN}^{-1} \Sigma$ $\text{cond}(P_{DN}^{-1} \Sigma) = O(1)$! indep. of h

$\rightarrow P_{NN}^{-1} \Sigma$ $\text{cond}(P_{NN}^{-1} \Sigma) = O(1)$! indep. of h

$\Rightarrow f_{DN}$ and f_{NN} will be INDEPENDENT of h !

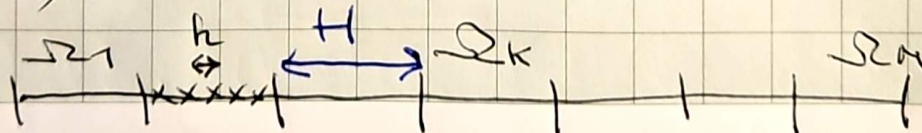
$\Rightarrow$ number of iterations to converge are independent of h !

$\Rightarrow$ [both DN and NN, for 2 subdomains, are OPTIMAL !



if you refine the mesh, the
 n° of iterations
 will not increase

Unfortunately, this is not the case if we use MANY subdom.



convergence rate will

degenerate with h, H

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NEITHER OPTIMALITY, NOR SCALABILITY

Orientation:

Solving 1 iteration of DN or NN on the FE system is equivalent to solving 1 iteration of Richardson type on the Schur complement system, that is:

$$\begin{cases} P(\vec{x}^{(k)} - \vec{x}^{(k-1)}) = b(\vec{x}_r - \sum \vec{x}^{(k-1)}) \\ \text{with } P = P_{DN} \text{ for DN or } P = P_{NN} \text{ for NN} \end{cases}$$

$$\Rightarrow \begin{cases} \text{DN} & \text{characterized by the precod. } P_{DN} \\ \text{NN} & \text{" " " " } P_{NN} \end{cases}$$

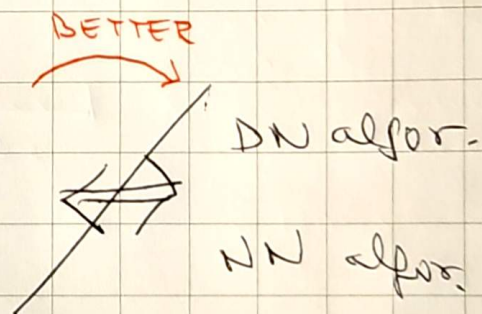
Now we know that there are MORE efficient iterative algorithms for $\sum \vec{x} = \vec{x}_r$.

For ex. Krylov, and, if Σ spd, CG

Why not using these algorithms INSTEAD of Richardson ??

$\Rightarrow$ Krylov, with $\begin{cases} P_{DN} \\ P_{NN} \end{cases}$

CG, with $\begin{cases} P_{DN} \\ P_{NN} \end{cases}$



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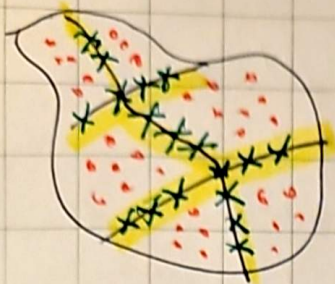
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MANY SUBDOMAINS

$$A \vec{u} = \vec{f}$$

FE System



$$\vec{u} = \begin{bmatrix} \vec{u}_I \\ \vec{u}_\Gamma \end{bmatrix} \begin{array}{l} \leftarrow \text{INTERIOR NODAL VALUES} \\ \leftarrow \text{INTERFACE NODAL VALUES} \end{array}$$

$$\Rightarrow \begin{bmatrix} A_{II} & A_{I\Gamma} \\ A_{\Gamma I} & A_{\Gamma\Gamma} \end{bmatrix} \begin{bmatrix} \vec{u}_I \\ \vec{u}_\Gamma \end{bmatrix} = \begin{bmatrix} \vec{f}_I \\ \vec{f}_\Gamma \end{bmatrix}$$

$$A_{II} = \begin{bmatrix} A_{II}^1 & & & \\ & A_{II}^2 & & \\ & & \ddots & \\ \emptyset & & & A_{II}^M \end{bmatrix} \quad \emptyset$$

$A_{II}^{(k)}$ is the stiffness FE matrix of Ω_k

$\Rightarrow$ Can construct the Schur complement system:

eliminate the internal nodes:

$$A_{II} \vec{u}_I + A_{I\Gamma} \vec{u}_\Gamma = \vec{f}_I \Rightarrow \vec{u}_I = A_{II}^{-1} \left[\vec{f}_I - A_{I\Gamma} \vec{u}_\Gamma \right]$$

Now:

$$A_{\Gamma I} \vec{u}_I + A_{\Gamma\Gamma} \vec{u}_\Gamma = \vec{f}_\Gamma$$

$$\Rightarrow -A_{\Gamma I} A_{II}^{-1} A_{I\Gamma} \vec{u}_\Gamma + A_{\Gamma\Gamma} \vec{u}_\Gamma = \vec{f}_\Gamma - A_{\Gamma I} A_{II}^{-1} \vec{f}_I$$

$$\underbrace{\left[A_{\Gamma\Gamma} - A_{\Gamma I} A_{II}^{-1} A_{I\Gamma} \right]}_{\sum} \vec{u}_\Gamma = \vec{X}_\Gamma$$



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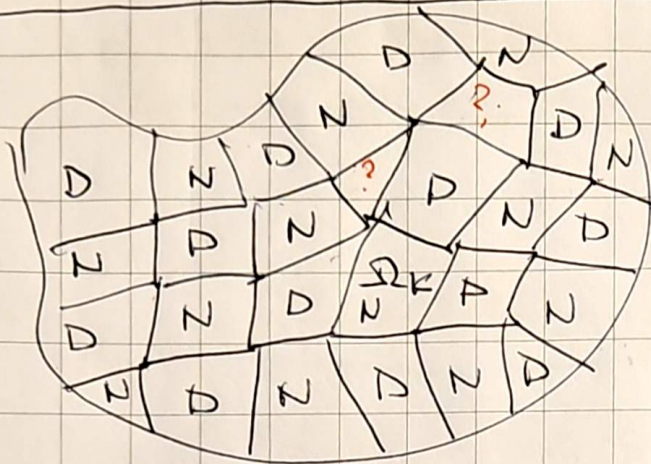
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$$(SC) \quad \boxed{\sum \vec{u}_r = \vec{X}_r} \quad \text{schur complement system}$$

Lesson learned in the case of 2 subdomains:

→ Apply Krylov or CG iterations
to (SC) using suitable generalization
of ~~DN~~ P_{DN} or P_{NN} preconditioners

Problems with DN in many subdomains



D subdomains
N subdomains
→ The DN partition
might not be
always possible.
(black & white subds)

Even worse in 3D!

→ DN algorithm is not practical
for many subdomains

→ We will restrict ourselves to NN algorithms!



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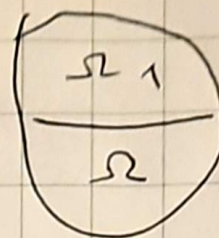
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$$P_{NN} = (\sigma_1 \Sigma_1^{-1} + \sigma_2 \Sigma_2^{-1})^{-1}$$

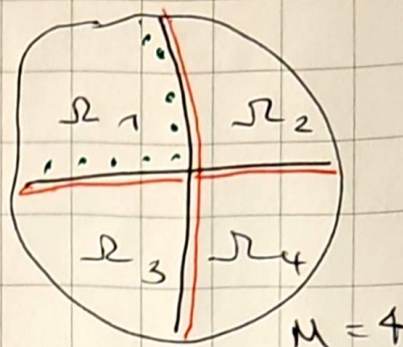
$\sigma_1, \sigma_2 > 0$ 2 subdomains



many subdomains:

$$P_{NN} = (\sigma_1 \Sigma_1^{-1} + \dots + \sigma_M \Sigma_M^{-1})^{-1}$$

local Schur complement
for Ω_1



P_{NN} operates on the whole skeleton Γ Red variables
 Σ_i operates only on the interfaces belonging to Ω_i
 local Green variables

$$P_{NN} = R_{\Gamma_1}^T D_1 \Sigma_1^{-1} + \dots + P$$

Green to Red
Prolongation matrix

$$P_{NN} = \left(R_{\Gamma_1}^T D_1 \Sigma_1^{-1} D_1 R_{\Gamma_1} + \dots + R_{\Gamma_M}^T D_M \Sigma_M^{-1} D_M R_{\Gamma_M} \right)$$

matrix of weights

$$D_k = \begin{bmatrix} \sigma_1^{(k)} & \emptyset \\ \emptyset & \sigma_d^{(k)} \end{bmatrix} \text{ coefficients (generalize } \sigma_1, \dots, \sigma_M)$$

Red to Green
Restriction matrix

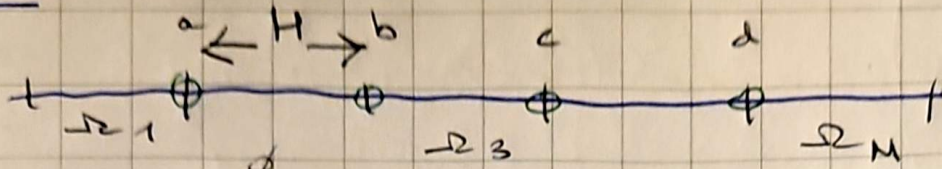


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1D

$$R_{\Pi_2} \cdot \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} b \\ c \end{bmatrix}$$

$$R_{\Pi_3} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R_{Π_k} = restriction matrix

$R_{\Pi_k}^T$ = prolongation matrix

Properties

$$\text{cond}(P_{NN}^{-1} \Sigma) \leq C H^{-2} \left(1 + \log \frac{H}{r} \right)^2$$

not scalable

not optimal

$M \uparrow \Rightarrow H \downarrow \Rightarrow H^{-2} \uparrow \Rightarrow$ conv. rate will become larger
 $\Rightarrow$ slower convergence

mesh finer $\Rightarrow h \downarrow \Rightarrow \log \frac{H}{h} \uparrow \Rightarrow$



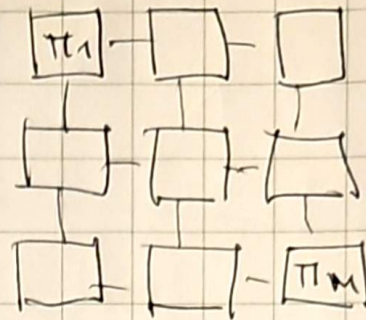
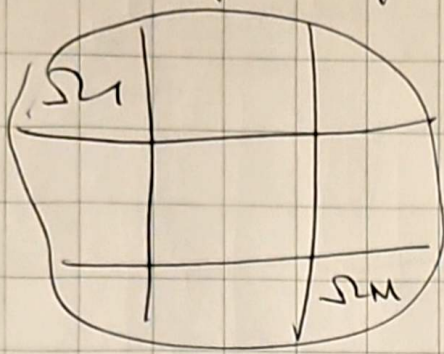
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localization by introducing a mechanism for a quick propagation of information

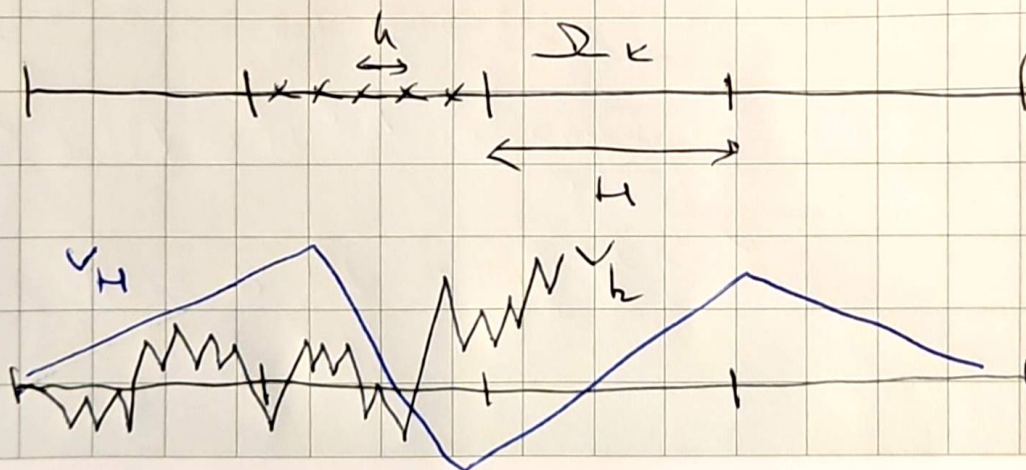


$$P_{NN}^{\text{CORRE}} = \left[\Sigma_H^{-1} + (I - \Sigma_H^{-1} \Sigma) (P_{NN}^{-1}) \cdot (I - \Sigma \Sigma_H^{-1}) \right]^{-1}$$

$$\Sigma_H = (R_P^T A_H^{-1} R_P)^{-1}$$

prolongation $\leftarrow$ $\leftarrow$ restriction

stiffness matrix
on the coarse mesh



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$$\text{cond}[(P_{NN}^{\text{worse}})^{-1} \Sigma] \leq c \left(1 + \log \frac{H}{h}\right)^2$$

virtually, this yields both
optimality & scalability

ex: $\text{cond}(\dots) \approx 2.07$

CG:

$$\beta = \frac{\sqrt{2.07} - 1}{\sqrt{2.07} + 1} \approx \frac{0.4}{1.4} \approx \frac{1}{3}$$

Strong scalability: Keep the pb. dimension N fixed and increase the # of processors $P \rightarrow$ reduce the computational time proportionally: N/P

Weak scalability

$\rightarrow$ increase the pb. size N and increase the number of processors P accordingly
 $\rightarrow$ keep the computational time constant



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